Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	17 October 2026
Project Name	Metro Ticket Generating System using ServiceNow
Maximum Marks	4 Marks

Technical Architecture

The architecture below describes the proposed ServiceNow-based metro ticket booking flow. The passenger uses the Service Catalog form to enter station and travel details; validation and fare logic prepare the request; Flow Designer coordinates booking processing; ticket information is stored; a QR-code digital ticket is made available; and Notifications confirm the booking. Exact implementation details should match the actual ServiceNow configuration.

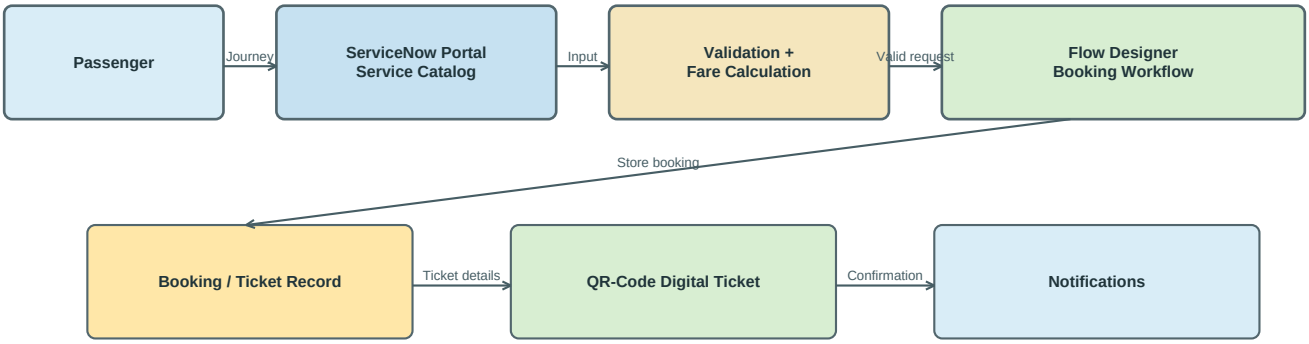

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Passenger booking form to select source/destination stations and enter travel details.	ServiceNow Service Catalog / Service Portal; configured catalog variables and UI components.
2	Application Logic-1	Validation of required fields and station selection.	ServiceNow Catalog configuration, Client Scripts / UI Policies where appropriate.
3	Application Logic-2	Automatic fare calculation based on configured station/fare rules.	Flow Designer logic or server-side scripts, depending on implementation.
4	Application Logic-3	Automate valid booking processing and ticket creation.	ServiceNow Flow Designer.
5	Database	Store booking details, fare, ticket reference, and ticket status.	ServiceNow platform table/records.
6	Cloud Database	Hosted platform data storage.	ServiceNow Now Platform cloud instance.
7	File / QR Ticket Output	Generate or store the QR-code representation and digital ticket details.	Configured QR-generation method and ticket record/attachment, as implemented.
8	External API-1	Send booking confirmation and ticket access information.	ServiceNow Notifications / email notification capability.
9	External API-2	No external integration specified in the core project description.	Not used in core scope; can be added if required.
10	Machine Learning Model	No machine-learning model is required for the described core booking workflow.	Not used in this phase; optional future enhancement only.
11	Infrastructure (Server / Cloud)	Run the booking form, workflow, data storage, and notifications.	ServiceNow Now Platform SaaS instance.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source	Use open-source web styling only if a custom portal widget is part of the implementation; the core workflow does not require an external open-source framework.	Bootstrap (optional, if used in the portal UI).
2	Security Implementations	Restrict booking and ticket data to authorized users using roles and access controls; protect traffic using platform security.	ServiceNow ACLs / role-based access; HTTPS/TLS.
3	Scalable Architecture	Use configurable station and fare data, reusable flow logic, and structured ticket records so requirements can grow.	ServiceNow tables, catalog configuration, and Flow Designer.
4	Availability	The service depends on the availability of the hosted ServiceNow instance and configured portal.	ServiceNow Now Platform service availability / applicable SLA.
5	Performance	Keep the form simple, validate fields efficiently, and avoid unnecessary server calls during booking.	Catalog configuration, Client Scripts / UI Policies where appropriate, and optimized Flow Designer logic.

References

1. C4 model – https://c4model.com/
2. IBM Cloud Architecture – https://www.ibm.com/cloud/architecture
3. AWS Architecture – https://aws.amazon.com/architecture
4. ServiceNow platform documentation should be consulted for the exact APIs and configuration used.